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Editorial essay

Risk environments and drug harms: A social science for harm reduction approach

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ABSTRACT

A ‘risk environment’ framework promotes an understanding of harm, and harm reduction, as a matter of ‘contingent causation’. Harm is contingent upon social context, comprising interactions between individuals and environments. There is a momentum of interest in understanding how the relations between individuals and environments impact on the production and reduction of drug harms, and this is reflected by broader debates in the social epidemiology, political economy, and sociology of health. This essay maps some of these developments, and a number of challenges. These include: social epidemiological approaches seeking to capture the socially constructed and dynamic nature of individual-environment interactions; political–economic approaches giving sufficient attention to how risk is situated differentially in local contexts, and to the role of agency and experience; understanding how public health as well as harm reduction discourses act as sites of ‘governmentality’ in risk subjectivity; and focusing on the logics of everyday habits and practices as a means to understanding how structural risk environments are incorporated into experience. Overall, the challenge is to generate empirical and theoretical work which encompasses both ‘determined’ and ‘productive’ relations of risk across social structures and everyday practices. A risk environment approach brings together multiple resources and methods in social science, and helps frame a ‘social science for harm reduction’.

Introduction

Drug harms are shaped by risk environments (Rhodes, 2002). There is a momentum of interest in understanding how the relations between individuals and environments impact on the production and reduction of drug harms. This is reflected by broader debates in the social epidemiology, political economy, and sociology of health, as well as HIV prevention (Gupta, Parkhurst, Ogden, Aggleton, & Mahal, 2008). This essay maps some of these developments.

One way to define a risk environment is to see it as the space – whether social or physical – in which a variety of factors interact to increase the chances of harm occurring (Rhodes, 2002, p. 88; Rhodes, Singer, Bourgois, Friedman, & Strathdee, 2005, p. 1027). This broad definition has led to envisaging a risk environment as comprising *types of environment* (physical, social, economic, policy) interacting with *levels of environmental influence* (micro, macro). Table 1 illustrates this simple heuristic, using selected examples relating to HIV risk and drug injecting (see also, Rhodes & Simić, 2005). This same logic implies an ‘enabling environment’ framework of social and structural change (Table 1). This is clearly not a model which purports to delineate causal pathways. And it is not theory in the positivist sense. But it is theoretical in its offering of a generative framework into which empirical and theoretical work might give primacy to *context* when understanding and reducing drug-related harm. Studies have unpacked the nature of risk environments in relation to varied drug-related harms, including overdose (Green et al., this issue; Moore, 2004), syringe sharing (Rhodes et al., 2003; Small, Kerr, Charette, Schechter, & Spittal, 2006; Strathdee et al., 2008), and sex work (Shannon, Kerr, et al., 2008; Shannon, Rusch, et al., 2008).

A ‘risk environment’ framework envisages drug harms as a product of the social situations and environments in which individuals participate. It shifts the responsibility for drug harm, and the focus of harm reducing actions, from individuals alone to include the social and political institutions which have a role in harm production. This essay considers different social science perspectives on the relationships between risk environments and drug harms. It serves to introduce this journal special issue.

Social science critiques of the relations between drug harms, individuals and environments can present daunting challenges from a pragmatics of harm reduction perspective. They include the increased recognition of the ‘nonlinearity’ of interactions in the difficult to delineate pathways between environments and drug harms, the problems posed by the inseparability of ‘levels’ of environmental influence, the seemingly daunting array of ‘factors’ produced by models of risk environment, and how to capture the dynamism of reciprocal relations between environments and individuals in which individuals at once ‘embody’ their risk

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Table 1
The risk and enabling environment: selected examples relating to HIV and drug injecting[a].

		Micro-environment	Macro-environment
Physical			
	Risk	Drug using, injecting and sex work locations	Drug trafficking and distribution routes
		Drug injecting in public spaces	Trade routes and population mobility
		Prisons and detention centres	Geographical population shifts and population mixing
	Intervention	Creating safer drug using sites (e.g. sharps disposal, lighting)	Changes to trafficking interdiction policies
		Developing supervised injecting facilities	Interventions at truck stops and train stations
		Prison-based harm reduction interventions	Cross-border interventions
Social			
	Risk	Social and peer group 'risk' norms	Gender inequalities and gendered risk
		Local policing practices and 'crackdowns'	Stigmatisation and marginalisation of drug users
		Community health and welfare service access and delivery	Weak civil society and community advocacy
	Intervention	Social network and peer-based interventions	Fostering collective actions in combination with policy changes
		Police partnership and training projects	Mass media and social marketing of harm reduction
		Developing low threshold accessible services for drug users	Strengthening civil society infrastructure and self-help
Economic			
	Risk	Cost of living and of health treatments	Lack of health service revenue and spend
		Cost of prevention materials	Growth of informal economies
		Lack of income generation and employment	Uncertain economic transition
	Intervention	Subsidised and free treatment	Increase investment in harm reduction relative to enforcement
		Distribution of free prevention materials	National health insurance schemes
		Micro-economic enterprise and employment schemes	Laws governing employment rights
Policy			
	Risk	Availability and coverage of clean needles and syringes	Public health policy governing harm reduction and drug treatment
		Programme-level policies governing distribution of materials	Laws governing possession of drugs
		Access to low-threshold and social housing	Laws governing protection of human and health rights
	Intervention	Scaling-up pharmacy-based syringe provision	Legal reform enabling the scaling-up of harm reduction
		Secondary syringe distribution programmes	Legal reform enabling the protection of drug user rights
		Hostel-based and housing neighbourhood development	National policy changes regarding public health strategy

[a] See also, Rhodes and Simić (2005).

environments and adapt them (Agar, 2003a; Duff, 2007; Fitzgerald, this issue). The emphasis of this work is theorising the risk environment, but *in order to act*.

Let us remind ourselves of how a risk environment approach enables harm reduction. First, it offers a critique on a tendency in public health and the behavioural sciences to emphasise harm as a primary determinant of individual action and responsibility. Models of individual-level change are self-evidently limited in their capacity to explain, encourage or sustain sufficient change to adequately reduce or prevent harm. This shifts the *focus for change* from individuals alone to the social situations and structures in which they find themselves. Of key interest here is the creation and enhancement of 'enabling environments' for harm reduction. Second, it offers a critique to resist *blame for harm* being directed toward affected individuals and communities, and shifts *responsibility for harm* to include the social and political–economic institutions which have a role in harm production. This explains its affinity with political–economic perspectives (see below). Third, a focus on *risk as socially situated* offers impetus to understanding how risk environments are *experienced* and *embodied* as part of everyday practices. This helps build ground-up or 'emic' understandings of risk subjectivity and environment (also see below). Fourth, and by extension, it implies the incorporation of harm reduction inside broader frameworks *promoting human rights* approaches to public health. By making visible the social, political and economic dimensions of risk and its regulation, a risk environment approach opens up opportunities for reducing the social suffering of which drug harms are a part.

Social epidemiology

It has been suggested that behavioural science is at a "crossroads" in public health (Glass & McAtee, 2006). This crossroads is characterised by the need for 'paradigm shifts' towards models and methods capturing the reciprocal relationships between environments and individuals. Considerable momentum has been created by repeated calls for a 'social epidemiology' (Diez-Roux, 2007; Krieger, 2001; McMichael, 1999; Susser & Susser, 1996), and for structural intervention approaches, including in HIV prevention and harm reduction (Blankenship, Friedman, Dworkin, & Mantell, 2006; Galea, Nandi, & Vlahov, 2004; Gupta et al., 2008; Poundstone, Strathdee, & Celentano, 2004). This places emphasis on how risk and harm at the population level is mediated by, or a product of, determinants that extend beyond proximate individual-level factors and their biological mediators.

Yet despite calls for "theory" (Krieger, 2001), and "better theory" (Glass & McAtee, 2006, p. 1651) in (social) epidemiology, talking of a 'paradigm shift' as imminent in applied public health is probably unrealistic. An emphasis upon "risk factorology" (Pearce & McKinlay, 1998) appears to endure. This is the preoccupation of giving primacy to studying proximate risk factors as determinants (Krieger, 2008). An emphasis on individual-level risk factors and behaviour change predominates harm reduction. Attempts to capture interplay between risk environments and health actions concentrate largely on delineating 'constraints' on 'volitional action'. Emphasis is given to narrow concepts such as 'self-efficacy' and 'readiness to change' within an epistemology of cognitive-based rational and planned action (Resnicow & Page, 2008). This is no paradigm shift. At the root of resistance to change is grasping the full implications of the idea that health and causation are *contingent*: "Epidemiology is being dragged, kicking and screaming, towards complexity and contingent causation" (Glass & McAtee, 2006, p. 1658).

Social epidemiology appears to hold great promise as a challenge to biomedical individualism in mainstream public health. An emphasis on how 'upstream' or 'distal' environmental factors integrate with 'downstream' or 'proximal' exposures at least de-emphasises those risk factors 'closest' to 'outcomes' which translate most directly to biological causes, even if social determinants are not seen as fundamental or decisive (Diez-Roux, 2007; Krieger, 2001). At the same time, a population focus concentrates on the distribution of harm in whole populations or societies rather than focusing on high risk individuals and their susceptibility to harm. This is because population health is neither the mere aggregation of individual risk factors nor the directly connected upstream determinants of proximate factors (McMichael, 1999, p. 893).

The effect of societal transitions offers an example. Societal transitions in Eastern Europe have been described as a risk environment linked to increased mortality through alcohol use and HIV infection through injecting drug use (Bobak & Marmot, 1996; Leon et al., 1997, 2007; McKee, 2002; Rhodes & Simić, 2005; Rhodes et al., 1999; Walberg, Mckee, Shkolnikov, Chenet, & Leon, 1998). The massive divergence in East–West European mortality and morbidity is not easily explained at the level of individual-level behaviours alone. In Russia, community disengagement, shifting social capital, and the erosion of traditional male roles followed political and economic transition, and these factors appear linked to the dramatic rise in male mortality attributable to excessive alcohol consumption in the 1990s (Leon et al., 1997; Walberg et al., 1998). Declines in life expectancy, and shifts in patterns of substance use, coincide with the uneven distribution of income, turbulence in labour markets, increased participation in criminal economies, and weakening civil society infrastructures (McKee, 2002; Walberg et al., 1998). Similar patterns have been observed in relation to the rapid diffusion of injecting drug use, with consequent HIV outbreaks (Laetitia, Carael, Brunet, Frasca, & Chaika, 2000; Rhodes & Simić, 2005; Rhodes et al., 1999).

Additionally, recent research links health contingencies to *social stress* in environments characterised by transition, disruption or discrimination. There is evidence linking macro-level traumas, such as terrorism, including the events of 9/11, with social stress and increased alcohol and drug use (Vlahov, Galea, Ahern, Resnick, & Kilpatrick, 2004), including some years after the event (Richman, Cloninger, & Rospenda, 2008). The experience of everyday social discrimination and inequality may be linked to health harm and substance use via stress (Gee, Spencer, Chen, & abd Takeuchi, 2007; Gee, Delva, & Takeuchi, 2007; Gebo, Keruly, & Moore, 2003; Harrell, Hall, & Taliaferro, 2003; Siahpush et al., 2006). Chronic exposure to social stress may impact *physiologically* through allostatic load (the 'wear and tear' of the body resulting from the accumulation of stressors), as posited for cardiovascular disease (McEwen, 1998). The risk environment is not merely an antecedent or mediator of biological process, but is internalised biologically (Diez-Roux, 2007; Glass & McAtee, 2006; Taylor & Repetti, 1997). Embodiment here then, is the physiological manifestation of the interface between biological evolutionary trajectory and the social-material world (Krieger, 2001, p. 672).

The examples of societal transition and terrorism are useful because they draw attention to health as the manifestation of *system* rather than aggregated individual-level effects, and because they emphasise the relevance of social forces which stretch above and beyond the local. This is important: "In our globally linked lives, identities come from all over the map, and from history and imagined futures as well" (Agar, 2003a, p. 981). The delineation of 'levels' of social organisation in epidemiology de-emphasises how risk contexts are made, in part, through their historical and global connections. Moreover, the dynamism of interactions between individuals and manifold factors in the risk environment across multiple levels tends to go missing in traditional epidemiological accounts. Age-old distinctions between 'distal' and 'proximal' in epidemiology collapse as events at any one level affect events at another, often without discernable intermediaries (Krieger, 2008). Levels of environmental influence coexist simultaneously, not sequentially, and we embody both at the same time. Risk exposures intermingle with ecosystems, economics, politics and history at "*every* level, macro to micro, from societal to inside the body" (Krieger, 2008, p. 227). This blending of interactions between different aspects of environment and individuals is not rational or linear in its effects as idealised in regression models of linear cause and effect, and neither is it random, but it is erratic and unpredictable:

> "Ecological models tend to draw from traditional systems theory, with its emphasis on control and equilibrium, when, in fact, a person-in-context is likely to be more about adaptation and change" (Agar, 2003a, p. 983).

This presents a daunting challenge for models delineating pathways to harm reduction (Friedman, Rossi, & Braine, this issue), and also perhaps explains the slow progression towards a social epidemiology of health. It shifts thinking towards an understanding of risk, and risk behaviour, as non-linear "quantum events" which are highly variable, difficult to predict, and "chaotic", wherein the results of interactions between factors are not only dynamic but "greater than the sum of their parts" (Resnicow & Page, 2008, p. 1382). A social epidemiology of drug harms needs to emphasise a "complex system" of non-linear (as well as linear) interactions occurring within and between individuals and their environments (Galea, Hall, & Kaplan, this issue). It accepts that pathways to individual-level behaviour change are "virtually infinite" (Resnicow & Page, 2008, p. 1382). It also entertains the idea that the 'risk factor' has limited use, perhaps even life-expectancy (Diez-Roux, 2007; Glass & McAtee, 2006). This is because any interaction occurring between individuals and environments constitutes, arguably, a moment of 'assemblage' which, critically, is *transformative* in its reciprocal effects, and thus difficult to delineate or predict.

In this issue, Galea et al. (this issue) make the case for embracing a "complex system" approach. Noting that "multi-level thinking has arrived" in drug use research, they note an emerging evidence-base exploring the reciprocal nature of interactions shaping drug harms (Agar, 2003a; Ahern, Galea, Hubbard, Midanik, & Syme, 2008; Bernstein, Galea, Ahern, Tracy, & Vlahov, 2007; Galea et al., 2003, 2004; Williams & Latkins, 2007). They illustrate their case through a simulation which uses 'agent-based modelling' to understand drug dependence. This work highlights the methodological challenges involved in bringing epidemiology into the complexity of the "real world", hitting home why it is fundamentally important to build epidemiology from the ground-up (Agar, 2003a, 2005). This work points to the potential for multi-level interventions in harm reduction (Fuller et al., 2007).

There has been concentrated effort to model relationships between geographic area and health outcomes (Diez-Roux, 2002). An evidence-base links 'neighbourhood effects' and drug harms (Cooper, Friedman, Tempalski, & Friedman, 2007; Galea et al., 2003; Latkin, Williams, Wang, & Curry, 2005; Maas et al., 2007; Williams & Latkins, 2007). Cooper, Bossack, Tempaslki, Des Jarlais, and Friedman (this issue) provide an illustration of such work. Noting that syringe exchange interventions and law enforcement activities are structural and spatial in nature, they explore methods to 'quantify' the geospatial drug injecting risk environment through relationships between drug harms, spatial access to syringe exchange, and exposure to drug-related arrests in New York City. The papers by Galea et al. (this issue) and Cooper et al. (this issue) add momentum to calls for a social epidemiology of drug harms, but as importantly, debate some of the methodological implications.

Political economy

An emphasis in social epidemiology on reducing health inequalities draws attention to an overlap with political–economic perspectives, at least in some accounts (Krieger, 2001, 2008). These consider how economic and political institutions produce and reproduce social and economic conditions which shape inequalities in health and access to services (Doyle, 1979; Navarro & Muntaner, 2004). For Krieger, social epidemiology extends the political economy of illness to include effects on biology, through the embodiment of social-material conditions (Krieger, 2005, 2008). Health, she notes, "cannot be divorced from considerations of political economy and political ecology" (Krieger, 2008, p. 223). This reflects parallel assertions in the social sciences that the pharmacology, use, and harms of drugs are "virtually meaningless outside their socio-cultural as well as political economic contexts" (Bourgois, 2003, p. 32), and that drug use is "the epiphenomenonal expression of deeper, structural dilemmas" (Bourgois, 1995, p. 319).

Studies document the political economy of harms as well as access to services linked to illicit as well as licit drug use (Agar, 2003b; Bourgois, 1998, 2003; Bobrova et al., 2006; MacDonald & Marsh, 2001; Musto, 1999; Singer, 2001, 2008; Wolfe, 2007). For example, Peretti-Watel, Seror, Constance, and Beck (this issue) show how patterns of tobacco smoking in France link with poverty, with differentials in rates of smoking on account of socio-economic condition widening over time. Nasir and Rosenthal (this issue) find that the socio-economic risk environment of urban slum areas ('*lorong*') in Makassar, Indonesia, structures the social context of initiation into drug injecting among men, in part through their pursuit of masculinity ('*rewa*'). In the absence of opportunities to accomplish *rewa* through income and socio-economic position, it is accomplished in alternative ways, including through gang membership and openness to risk involving drug use.

Ciccarone and colleagues have explored how global and political–economic shifts in heroin markets act as structural forces affecting local drug use risk environments (Ciccarone, 2005; Ciccarone & Bourgois, 2003). In this issue, he shows how shifts from Asian towards Colombian and Mexican heroin in the U.S. heroin market followed the diversification of Colombian drug supply from cocaine to heroin as a paradoxical effect of U.S. interdiction efforts, also facilitated by neo-liberal economic reforms in the region. Structural shifts in the heroin marketed in the U.S. impacts directly and differentially on the drug injecting risk environment in different parts of the country, as the source and type of heroin available shapes how the drug is used, and consequently, the form and intensity of health risks experienced.

A tenet of political–economic perspectives is that they posit socio-economic condition as rendering *particular* sectors of the population vulnerable to harm. Let us take crack as an example. Agar notes how the crack epidemics in the early 1980s in the U.S. were fuelled by a cocaine glut coinciding with the decline of urban manufacturing industries, weakening welfare support services, and widening income disparities. Its diffusion centred on "impoverished" urban African-American communities, reproducing a "racist self-fulfilling prophecy" (see also Reinarman & Levine, 1995). Bourgois, too, sees trends in crack use in this period as a function of "inner city apartheid" (Bourgois, 2003, p. 32), disproportionately affecting the exploited who suffer from intense and systematic discrimination, including racial discrimination. Crack thus represented "a good and rare business opportunity in a time of extreme social and economic decline" (Agar, 2003a, p. 20). It was a "rational career choice" towards realising the American Dream (Adler, 1995).

Drug harms are thus a feature of the political economy of social suffering. Singer proposes the concept "oppression illness" (Singer, 2004, p. 17). This is the "product of the impact of suffering from social mistreatment"; a type of "stress disorder", where the source of stress is "being the object of widespread and enduring social discrimination, degradation, structural violence and abusive derision", whether overt or hidden (Singer, 2004, p. 17). It is the process through which an oppressive social environment is incorporated into the everyday practices of those subjected to multiple subordinations (Friedman et al., 1998). As noted, this idea is corroborated by epidemiological studies linking chronic health conditions, as well as substance use, to social discrimination and social stress (Gee, Spencer, Chen, et al., 2007; Gee, Delva, et al., 2007; Richman et al., 2008; Siahpush et al., 2006). Peretti-Watel et al. (this issue) suggest that tobacco smoking is a means of "coping" with "socio-economic stress" and "stressful environments". Thus, social conditions find their expression at the level of the individual, including in terms of psychological and emotional harms, self-blame, reduced sense of self-worth, and crucially, risk behaviour (Parker & Aggleton, 2003; Rhodes et al., 2005). Singer therefore sees drug use as a form of "*self-medication*" for oppression illness (Singer, 2004, p. 17). Oppression illness "pressures sufferers to seek relief", where drug use is an "action-oriented culture" emphasising "gratification", "pain intolerance", "chemical intervention" and a "solution" (Singer, 2001, p. 205).

Of course, one of the most visible structural mechanisms perpetuating social suffering is the criminal justice system. A large literature points towards policing practices and fear of the criminal justice system contributing to drug harms (Davis, Burris, Metzger, Becjer, & Lunch, 2005; Friedman et al., 2006; Kerr, Small, & Wood, 2005; Miller et al., 2008; Rhodes et al., 2003; Shannon, Kerr, et al., 2008; Shannon, Rusch, et al., 2008; Werb et al., 2008). Policing practices targeting the vulnerable act as institutionalised expressions of social and moral regulation, manifest in everyday technologies of surveillance (Rhodes et al., 2006, 2007) through to excessive force (Cooper, Moore, Gruskin, & Krieger, 2005; Rhodes, Simić, Baros, Žikić, & Platt, 2008). Prisons are built expressions of risk environment, including for HIV, and like most other forms of criminal justice intervention, disproportionately affect minority populations (Bourgois, 2003; Galea & Vlahov, 2002; Lemelle, 2002). In some countries, it is very difficult to explain contemporary drug epidemics and the harms they cause without acknowledging what Bourgois refers to as the iatrogenic effects of "carceral drug policy" (Bourgois, 2003, p 36).

The internalisation of social suffering reproduces a cycle of risk production in which those marginalised can become complicit, including unconsciously, in their ongoing structural subordination (Kleinman, Das, & Lock, 1997). In this way, symbolic and structural violence is reproduced through everyday interactions *between drug users*—for example, as shame (Rhodes et al., 2007), stigma (Simmonds & Coomber, 2009), and gendered violence (Maher, 1997), as much as through interactions between drug users and health systems (Bourgois, 2000; Radcliffe & Stevens, 2008). The structuring of drug harms through gendered violence illustrates well how social suffering functions through everyday micro-level practices which 'normalise', and thus 'render invisible', processes of inequality (Bourgois, Prince, & Moss, 2004; Maher & Daly, 1996; Rhodes & Quirk, 1998).

Risk perceptions and actions emerge within, and reflect, social and cultural responses to political–economic conditions. Returning to the example of societies in transition, Pilkington and Sharifullina (this issue) draw upon ethnographic research among young people in Vorkuta, a city of the Russian far north. They are interested to explore links between macro-economic transitions characterised by deindustrialisation and social capital at the micro-level. Whereas societies in transition are usually depicted as lacking in social capital (McKee, 2002), and most studies of drug use posit social networks as protective against risk (Friedman et al., 2007), they characterise young people's social networks as driven by mutual exploitation and mistrust centring around informal economic practices in which drugs play a key role. Social network practices are shaped culturally by openness to risk. They are shaped structurally by economic transformations which have disrupted normative capital and economic relations, which once visible become open to question, in turn enabling the "mutual extraction" of capital in new ways at the micro-level. Friedman et al. (this issue) also investigate links between changing political–economic conditions and normative regulations, proffering a pathways model of risk environment to capture how societal transitions might impact on drug harms.

Lived experience

Political–economic approaches have been criticised from a pragmatics of harm reduction perspective because they imply the need for large-scale social transformations in how societal systems function. As Agar notes, "saying that poverty is a risk factor for drug use does not help much. What is the clinician to do, tell the person to stop being poor?" (Agar, 2003b, p. 983). This cautions against models of risk environment which give inadequate account of how structural forces can impact *differentially* in lived experience at the local level (Duff, 2007). They have also been criticised for *underplaying agency*; for positioning individuals as largely passive in their complicity to 'structural determinants' (Fitzgerald, this issue).

Risk environments may constrain how agency is enabled, but they are at once also a product and adaptation of agency. This recalls our discussion above regarding how social epidemiology is challenged to capture the dynamism of the reciprocity of individual-environmental interactions. Risk environments feature in a process of what Giddens called "*structuration*" (Giddens, 1984). Structuration is the reciprocal and adaptive effects of relations between social structures and individuals. Structure is not 'external' to individuals, for the "constitution of agents and structures are not two independently given sets of phenomena, a dualism, but represent a duality" (Giddens, 1984, p. 25). This means that "social systems are both medium and outcome of the practices they recursively organise" and that structure is "not to be equated with constraint but is always both constraining and enabling" (Giddens, 1984, p. 25).

Drawing upon a qualitative case study in the UK, Fletcher, Bonell, Sorhaindo, and Rhodes (this issue) explore relationships between an inner-city school risk environment and cannabis use. They show that an important characteristic of the school environment is *insecurity*, which structures the social organisation of peer groups and identities in relation to cannabis. Cannabis use and related 'safe' identities emerge as *solutions to risk* in this particular risk environment, especially among the most disengaged students with least resources. Recalling the observation that drug use can function as a form of 'medication' in response to social suffering and stressful environments (Peretti-Watel et al., this issue; Singer, 2004), cannabis use is described here as a form of "anti-stress self-medication". The authors also describe their case study as "structuration in action". They argue that the inner city school risk environment at once reflects and structures patterns of drug use though social conditions which constrain some actions while enabling others. Structural inequalities regarding ethnicity and socio-economic status are reproduced in this setting through drug use. Even while collective social identities have weakened, and risks have become increasingly 'individualised' and 'globalised' (Beck, 1992; Giddens, 1991), locality, peer group and structured inequalities remain important to lived experiences of drug use for young people in the UK (Furlong & Cartmel, 1997; MacDonald & Marsh, 2001; Mitchell, Crawshaw, Bunton, & Green, 2001).

A risk environment approach thus explores how experiences of risk are socially situated. Moore, for example, considers overdose (Moore, 2004). He shows that discourses of overdose prevention informed by the practices of epidemiology and behavioural science articulate particular subjectivities and technologies of the body as well as images of environment. These emphasise rational decision-making on the part of drug users to enact strategies such as avoiding poly-drug use or injecting alone and testing the strength of heroin prior to use. Overdose prevention discourses also assume a social context characterised by "stability and orderliness" (Moore, 2004, p. 1550). His ethnography shows how everyday practices are located in an *alternative* cultural logics, produced by an unpredictable environment, in which multiple risks compete for risk management attention, and in which manifold social forces combine to undermine or disrupt public health risk rationality. This work hits home that risk is relative; that rationalities about risk are situated socially and culturally, as well as politically (Douglas, 1992; Lupton, 1999; Rhodes, 1995, 1997).

Understanding the socially situated nature of risk rationalities draws attention to how the discourses and practices of public health inculcate subjectivity and everyday life in relation to risk (Moore & Fraser, 2006). It is argued that neo-liberal forms of public health governance in Western societies characterise risk as primarily a responsibility of autonomous individuals who have a duty to self-care (Peterson, 1997; Rose, 1996). Through the technologies of epidemiology and behavioural science, risk becomes a core stratagem of regulation through increased emphasis upon surveillance of *risk factors* to illness (Armstrong, 1995; Castel, 1991). Emphasis accordingly shifts from clinic-based treatments of concrete individuals exhibiting illness to the *anticipation of harm* through risk factor epidemiology at the population level, involving the regulation of risk acceptability against a population 'norm', and strong emphasis placed on individuals through health promotion to act responsibly by avoiding risk for their own and the public good (Bunton & Burrows, 1995). Harm reduction discourses act as forms of 'bio-power' in the social regulation of danger emanating from drug use and drug users (Bunton, 2001; Valverde, 1998).

One example concerns methadone treatment, a technology of harm reduction supported by an impressive evidence-base (Farrell, Gowing, Marsden, Ling, & Ali, 2005; Gowing, Farrell, Bornemann, Sullivan, & Ali, 2006). Using ethnography, Bourgois argues that methadone "represents the state's attempt to inculcate moral discipline into the hearts, minds and bodies of deviants who reject sobriety and economic productivity" (Bourgois, 2000, p. 167). He sees methadone treatment as 'biopower' at work. It acts as a medical technology for regulating pleasure and economic productivity by substituting an addictive and euphoric drug, heroin, for a more addictive less euphoric medicine, methadone, producing "docile" and "disciplined" patients. In contrasting methadone substitution with prescribed heroin treatments, Bourgois shows us that ethnography of lived experience can reveal, as well as *challenge*, the assumed objectivity of positivistic knowledge regarding risk regulation. Other critiques have focused on hepatitis C and HIV prevention (Fraser, 2004; Moore & Fraser, 2006), overdose prevention (Moore, 2004), and drug consumption rooms (Fischer, Turnbull, Poland, & Haydon, 2004). While criminal justice interventions represent acts of state discipline for those who require 'correction', public health technologies are more subtle forms of risk management incorporated through everyday active citizenship. This is the "bad cop" and "good cop" of drug user regulation (Bourgois, 2000). It shows the potential of a 'risk governmentality' approach to understanding everyday risk environments (Foucault, 1979; Lupton, 1999).

In emphasising harm as an exigency of liberal government and power-holding institutions, governmentality perspectives may also underplay agency, especially acts of resistance. A recent study extends a more nuanced analysis of the regulatory practices of methadone (Fraser & Valentine, 2008). This notes how the meanings of methadone are interpreted *variably* as well as *co-produced* through the everyday practices of methadone treatment subjects. Drawing on an ethnographic case study of drug dealing risk environments, Fitzgerald (this issue) likewise cautions against models of risk environment which perpetuate dichotomous models of 'structure' and 'agency'. He emphasises risk environments as embodied socially through everyday practices, and positions drug dealers as "*active participants*" in their local drug economy, and not merely as the "effects of structural oppression".

This gives a slightly different emphasis to understanding risk environment through lived experience. It sees structural and cultural influences emerging out of their 'incorporation' into embodied subjectivity, and in turn into *everyday practice* (Bourdieu, 1977; Lash, 2000). It shifts attention to the '*embodied subject*' of risk environment at a *local level*. Rather than delineating a composite of interacting macro forces acting upon individuals from outside, it sees an 'assemblage' of manifold interactions between individuals and environments being 'materialised' at the level of specific local contexts (Duff, 2007). This emphasis on the logics of everyday practice draws upon the theorising of Bourdieu. Key here is his notion that the structural and cultural conditions which make up particular environments produce particular kinds of *habitus* – socially acquired dispositions in relation to practices, habits, tastes and behaviours – which are reproduced iteratively, and often unconsciously, through everyday practices (Bourdieu, 1977, 1990). Risk environments are thus 'made' through day-to-day actions at the local level, and it is here that specific local differences *within* as well as *between* structural factors (such as class, gender, and ethnicity) can be accounted for. Whereas structuralism encourages an unproblematic acceptance, and thus reproduction, of the assumption that drug use is simply structured by socio-structural transformations, post-structuralist approaches encourage a focus on how class, gender and other structural factors are lived through *participation* in drug use and local drug economies (Maher, 1997; Vitellone, 2004, p. 146).

Lastly, a primary concentration on risk may undermine discovery of the ways in which environments can be capacitated to resist drug harms. Taking the city as his focus, Duff (this issue) explores urban drug use contexts for their potential to act as 'enabling environments' for harm reduction. He interprets this as more than the enablement of specific harm reduction measures such as syringe exchange (Moore & Dietz, 2005; Rhodes, 2002), to include how environments support *cultures* of resilience to risk. He identifies these as founded largely on social network and other relations of affect which generate social capital, informal support, solidarity and belonging. Friedman and colleagues have previously suggested the concept of '*intravention*' to describe how social networks in a neighbourhood mobilise collectively to create 'community resiliency', including in the face of drug and wider political–economic harms (Friedman et al., 2007). Together, these authors urge us to recognise that environments exhibit relations of risk *and* enablement, of disadvantage *and* capacity.

Risk environment: a social science for harm reduction

A risk environment framework promotes an understanding of harm, and harm reduction, as contingent upon social context. It is not agnostic in relation to arguments about 'fundamental causes' of harm, for it emphasises the primacy of social and material production. Yet notions of risk environment are articulated differently in social scientific accounts of drug harms. This essay has mapped some of these differences, summarised as Table 2.

Whilst entertaining post-positivism (Krieger, 2001, 2008), social epidemiology is challenged to capture the socially constructed dynamic nature of environment and its incorporation into everyday practices (Diez-Roux, 2007; Galea et al., this issue). These are, of course, fundamentally epistemological challenges, and not only methodological. A social epidemiology of drug harms needs to be approached from the ground-up as well as treated iteratively, as outlined in recent descriptions of 'qualitative epidemiology' (Agar, 2003a). Structuralist and political–economic perspectives give primacy to 'social determinants', but sometimes insufficient attention to how these are incorporated in different local contexts, and often underplay agency in this process. The 'totalising' claims of political–economic explanations of harm are taken by some to stultify community-level actions given a tendency to champion large-scale macro-social change. This again reinforces the need for emic perspectives on how risk experience and rationality is differently socially situated. While critiques emphasise public health and harm reduction discourses as sites of governmentality, inculcating risk subjectivity and everyday life, some of these also paint an overly passive picture of agency, and sometimes an overly dichotomous description of social environments 'determining' individuals (Fitzgerald, this issue). Moreover, illuminating the omnipresence of 'biopower', and harm reduction as one such site of governmentality, should not act as rationale for not acting (Keane, 2003; Moore, 2004). A focus on the 'embodiment' of risk environment into everyday practices offers a way to understand how structural forces might be co-produced differentially (Duff, 2007), but an emphasis on micro social relations inevitably underplays broader articulations of culture or context which make up all that operates above, behind and before the local (Agar, 2003a).

The challenge then is to generate empirical and theoretical work which encompasses both 'determined' and 'productive' relations of risk across structures and everyday practices. This challenge inevitably invites competing theoretical perspectives on risk environment—from weak to strong constructionist interpretations of risk as 'real' or 'imagined'; from models of structural determinism to the local production of risk subjectivity; and from methods modelling probable causal pathways to ethnographic stories of risk experience (Table 2). The objective here is not to attempt to overcome differences in varied visions of risk environment, for these differences are beyond resolution, and the relevant task is to consider what can be learnt from bringing these differences together.

This returns us to a statement earlier made: that theories of risk environment are there *in order to act*. Following arguments elsewhere (Nettleton & Bunton, 1995), Moore has used a risk environment approach to argue for 'an anthropology *for* heroin overdose prevention' as distinct from 'an anthropology *of* heroin overdose prevention' (Moore, 2004, p. 1548). What has been outlined here is an argument for using a risk environment approach to frame a *social science for harm reduction*. Moore acknowledges that advocating 'a risk environment and enabling approach' is inevitably a stratagem of regulation, for it constructs drug users and environments in particular ways, but that it promises "new forms of governmentality which might produce *less* social suffering" (Moore, 2004, p. 1548). A framework of risk environment is a call to mobilise resources in social science towards reducing drug-related social suffering. In doing this, it invites the bringing together of political–economic and socio-cultural research paradigms with shifts towards 'eco-social' approaches in public health epidemiology.

In illustrating the variant ways in which research on risk environments can proceed, this essay has argued that 'contingent causation' is at the heart of a risk environment approach. This

Table 2
Overlapping theoretical perspectives of risk environment in relation to drug harm[a].

	Empirical focus	Epistemological assumptions	Examples in drugs research
Social epidemiology (positivism/post-positivism)	What drug harms exist, how are these distributed in populations, and how are they the product of individuals interacting with their environments?	Risk and harm is an outcome, embodied biologically, and mediated through environmental processes, the distal and proximal causes of which are studied.	Bernstein et al. (2007), Cooper et al. (2007, this issue), Galea et al. (2003, 2004, this issue) and Poundstone et al. (2004)
Political economy (structuralism)	How are drug harms structurally determined by political–economic conditions (such as inequalities relating to income, class, gender, ethnicity)?	Risk and harm is an outcome inseparable from environmental processes and social–economic conditions. Social–economic condition also shapes cultural responses to risk.	Bourgois (1995), Bourgois (2003), Friedman et al. (1998, 2007), MacDonald and Marsh (2001) and Singer (2001, 2008)
Situated rationality (structuralism/post-structuralism)	How are drug harms constructed by socially and culturally situated knowledge of risk acceptability and rationality, as well as situated by everyday dangers which compete for risk management attention?	Risk and harm is viewed as a product of knowledge that is historically, socially and culturally produced. There are thus competing rationalities of risk. Responses to risk are shaped by social situation and social structures.	Bourgois (1999), Maher (1997), Mayock (2002), Moore (2004), Rhodes (1995, 1997) and Rhodes et al. (2003)
Governmentality (post-structuralism)	How do discourses and practices regarding drug harms construct related subjectivity and experience?	Risk and harm is a product of knowledge that is historically, socially and culturally produced. Discourses of risk are forms of normative regulation.	Bourgois (2000), Fischer et al. (2004), Moore (2004), Moore & Fraser (2006) and Valverde (1998)
Logics of practice (post-structuralism)	How do social conditions relevant to producing drug harm become incorporated into, and reproduced through, everyday practices and subjectivity?	Risk and harm is a product of knowledge that is historically, socially and culturally produced, which shapes everyday habits and practices, which in turn frames individual agency.	Duff (2007), Fitzgerald (this issue), Fraser & Valentine (2008) and Vitellone (2004)

[a] Informed by Lupton (1999, p. 35).

challenges future approaches from becoming overly structurally determined, to make problematic their distinctions between 'structure' and 'agency', 'distal' and 'proximal', and 'macro' and 'micro', and to question a reliance upon simple linear pathways of cause and effect. Interventions which target the social conditions producing drug harms may be more effective than interventions targeting specific behaviour changes among drug users, *even if* these social conditions are not easily translated into specific epidemiological causes or risk factors (Glass & McAtee, 2006). Most importantly, redistributing responsibility for harm and its reduction as something *shared* between individuals and social–economic structures is a force of resistance to neoliberal versions of risk rationality emphasising individual responsibility (and *blame*). This is why a risk environment approach is a social science for harm reduction which acts first to reduce social suffering.

Conflicts of interest

None to declare.

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